

Technical Report: Shelf-Life-Aware Buffer Tuning Results

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Executive Summary

This report documents the tuning of per-item safety stock buffers in the Flow-POS forecasting cost model. After identifying that the flat 30% waste penalty overstates costs for long-shelf-life items, three buffer adjustment strategies were tested through grid search. The best configuration achieves a **13.5% total cost reduction** (5,133,764 to 4,440,012 DKK) compared to the autoresearch baseline.

Background

The autoresearch best model (SoftProbModel) uses a flat safety buffer of 1.30x applied uniformly to all items. Initial evaluation showed that applying shelf-life-aware waste fractions alone (without changing predictions) reduces cost by 11.7%. The question was whether per-item buffer adjustments could yield additional improvement.

Previous Approach (Failed)

The initial smart buffer implementation used newsvendor theory: computing a per-item critical ratio from the shelf-life-adjusted waste fraction, converting to a z-score via inverse normal CDF, and scaling by the coefficient of variation. This failed because:

- Frozen items (effective waste near 0%) produce critical ratios approaching 1.0
 - `norm.ppf(0.999...)` returns infinity or near-infinity values
 - Even with clamping, the buffers were wildly inconsistent across items
 - Result: 5,690,556 DKK (10.2% worse than flat buffer)
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Tuning Methodology

Three buffer strategies were evaluated via grid search over the same model predictions:

1. **Uniform buffer sweep** — single buffer value for all items (1.10 to 1.40)
2. **Smooth adjustment** — $\text{buffer} = \text{base} + \text{scale} \times (\text{eff_waste} - 0.3) / 0.3$, clamped to [1.05, 1.50]

3. **Log scaling** — $\text{buffer} = \text{base} + \text{scale} \times \log(0.3 / \text{eff_waste})$, clamped to $[1.05, 1.50]$

The log scaling approach maps waste fractions smoothly: - High waste (perishable, $\text{eff_waste} = 0.3$): $\text{buffer} = \text{base}$ (no change) - Low waste (frozen, $\text{eff_waste} = 0.023$): $\text{buffer} = \text{base} + \text{scale} \times 2.57$ - Very low waste (ambient, $\text{eff_waste} = 0.007$): $\text{buffer} = \text{base} + \text{scale} \times 3.77$

Results

Top 10 Configurations

Rank	Strategy	Config	Total Cost	Waste	Stockout
1	log	base=1.30, scale=0.20	4,440,012	1,903,964	1,690,698
2	log	base=1.30, scale=0.15	4,443,106	1,900,763	1,694,895
3	log	base=1.25, scale=0.20	4,447,825	1,791,038	1,771,191
4	log	base=1.30, scale=0.10	4,448,123	1,897,171	1,700,634
5	log	base=1.25, scale=0.15	4,452,584	1,787,614	1,776,647
6	log	base=1.25, scale=0.10	4,459,105	1,783,794	1,783,541
7	log	base=1.30, scale=0.05	4,468,313	1,893,015	1,716,865
8	log	base=1.20, scale=0.20	4,468,881	1,679,895	1,859,324
9	uniform	buf=1.40	4,513,827	2,120,620	1,595,472
10	uniform	buf=1.30	4,556,010	1,888,285	1,778,483

Cumulative Improvements

Stage	Cost	vs Baseline
Autoresearch baseline	5,133,764 DKK	—
Shelf-life-aware cost model	4,556,010 DKK	-11.3%
+ Log buffer (base=1.30, scale=0.20)	4,440,012 DKK	-13.5%

Analysis

The log scaling strategy dominates all other approaches. Key observations:

1. **Log scaling consistently outperforms** — all top 8 configurations use log scaling. The smooth, monotonic mapping of waste fraction to buffer adjustment avoids the discontinuities that plagued the newsvendor z-score approach.
2. **Higher scale values help** — scale=0.20 (the maximum tested) consistently produces the best results. This means the model benefits from more aggressive differentiation between perishable and non-perishable items.
3. **Base buffer matters less** — base=1.25 and base=1.30 produce nearly identical results at the same scale. The scale parameter (how much we differentiate) matters more than the base level.
4. **Uniform buffers plateau at buf=1.40** — increasing beyond 1.40 would add waste cost faster than it reduces stockout cost. The per-item log approach outperforms even the best uniform buffer.
5. **Waste-stockout tradeoff shifts** — the best log config (4,440,012) has slightly higher waste cost (1,903,964) than the uniform buf=1.30 config (1,888,285) but significantly lower stockout cost (1,690,698 vs 1,778,483). The model orders more of long-shelf-life items (cheap overstock) while maintaining tighter control on perishables.

Winning Configuration

```
buffer = max(1.05, min(1.50, 1.30 + 0.20 * log(0.3 / effective_waste_fraction)))
```

Item Type	Effective Waste	Buffer
Fresh salad (shelf life 1d)	0.300	1.30 (no change)
Refrigerated chicken (5d, gap 3d)	0.180	1.38
Frozen beef (90d, gap 7d)	0.023	1.61 → clamped to 1.50
Bottled beer (365d, gap 9d)	0.007	1.83 → clamped to 1.50

The 1.50 cap prevents over-ordering even for near-zero-waste items.

Recommendation

Deploy the log-scaled buffer with base=1.30, scale=0.20 as part of the shelf-life-aware cost model. This provides a 2.5% additional improvement over cost-model-only changes, at negligible computational cost (one log + clamp per item per prediction).

The combined approach (shelf-life cost model + log buffers) achieves **-13.5% total cost reduction** versus the autoresearch baseline, equivalent to approximately **693,752 DKK in savings** on the test set.

Files

File	Description
flowpos_forecasting/autoresearch/evaluate_shelf_buffer.py	Upstream evaluation script
flowpos_forecasting/data/df16/classifications_shelf_life.csv	Shelf-life classification data

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